Designing a Self-Healing Ops Platform: Telemetry, RAG, and Confidence-Gated AI Agents

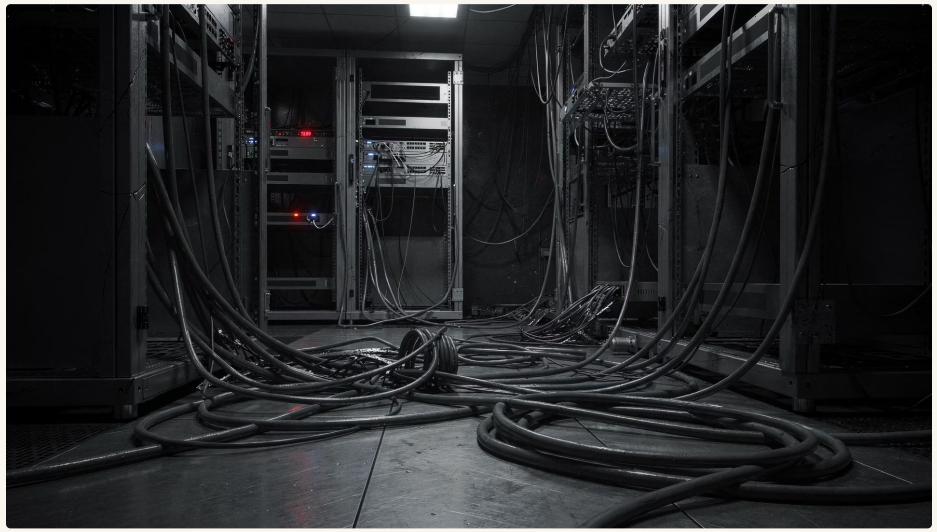
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Conf42 Platform Engineering 2026



The Problem: Silos That Keep Ops Reactive



Enterprise IT platforms layer monitoring, endpoint management, and ITSM tools that operate in isolation. The result: support teams stay permanently in firefighting mode.

Fragmented Signal

Telemetry, tickets, and asset data never merge into a unified picture.

Preventable Demand

Up to **40% of Level-1 requests** stem from issues existing telemetry could anticipate.

No Autonomous Action

Humans remain the only decision-makers, even for repeatable, low-risk fixes.



Introducing ADOF

The **Autonomous Digital Operations Framework** is a conceptual six-layer reference architecture that transforms reactive IT support into a proactive, governed, and continuously learning operations platform.

Unified Signals

One telemetry layer across all tools

AI-Augmented Knowledge

RAG connects agents to enterprise context

Specialized Agents

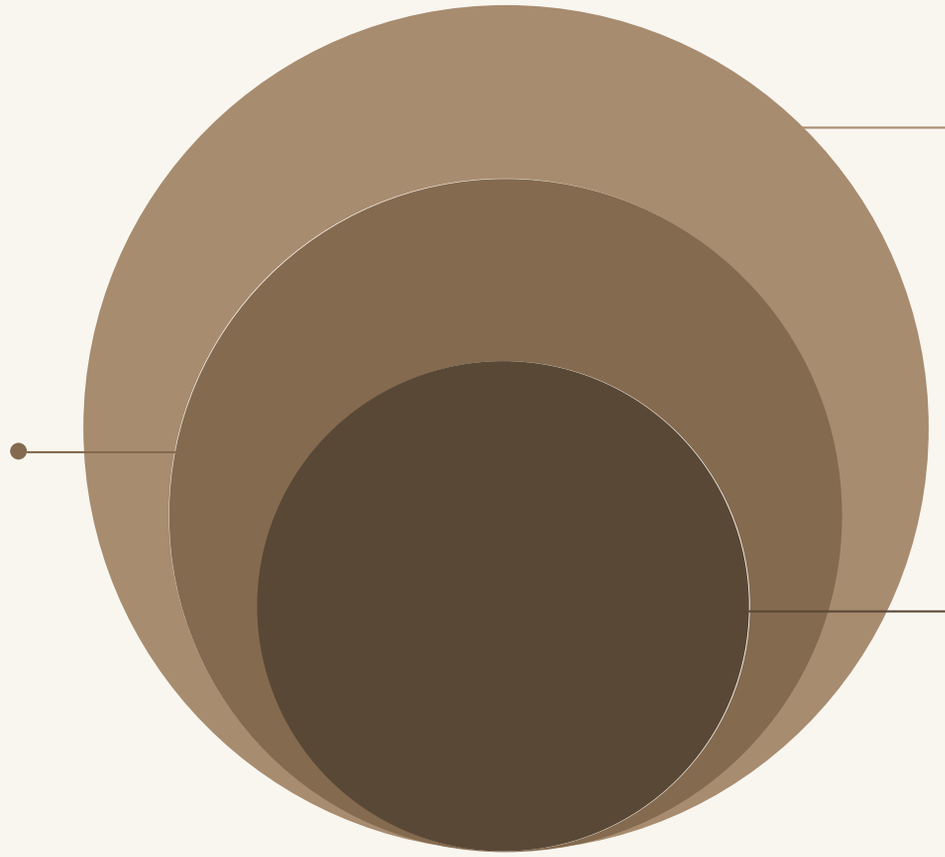
Six agents, each with a distinct responsibility

Confidence-Gated Action

Every automated fix is scored before execution

ARCHITECTURE DEEP DIVE

The Six-Layer Stack



Outer: Learning Loop

Capture outcomes for continuous improvement

Layer: Agent Orchestration

Coordinate agents and workflows

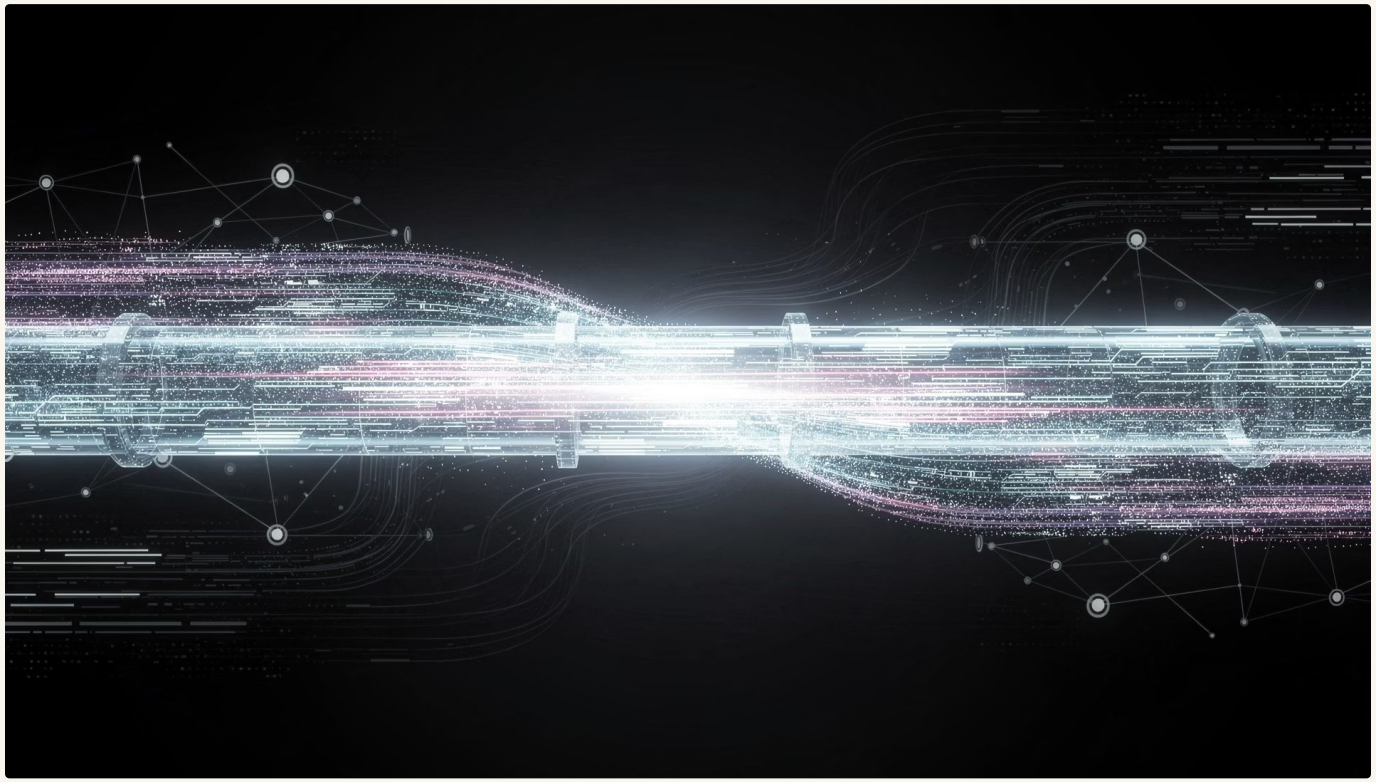
Core: Signal Unification

Aggregate and normalize raw telemetry

Each layer has a single responsibility. Together they form a closed-loop system from raw telemetry to governed autonomous action and continuous improvement.

LAYER 1

Signal Unification: One Truth for All Data



ADOF's foundation is a **unified signal layer** that ingests and correlates endpoint telemetry and ITSM data into a single, queryable stream.

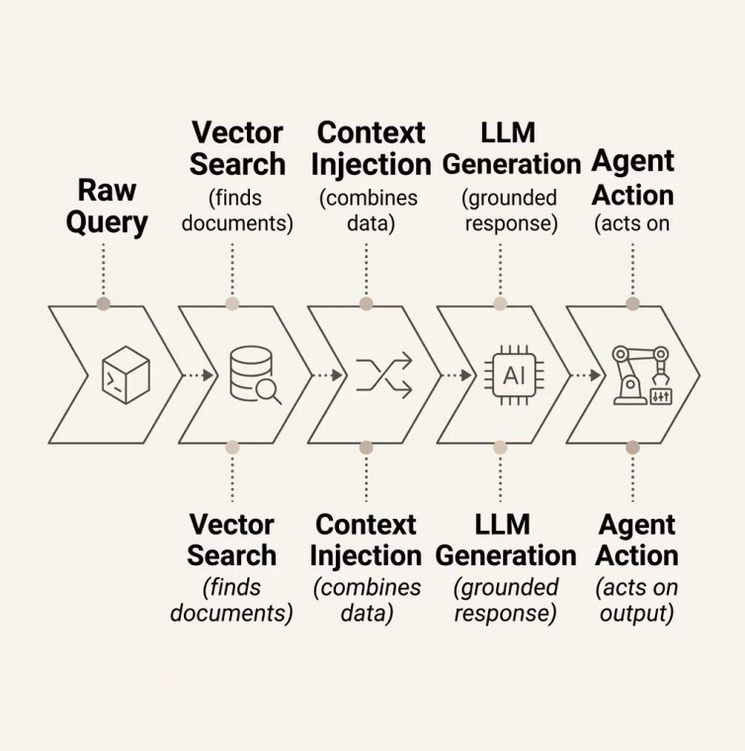
- **Endpoint telemetry**: CPU, memory, disk health, application crashes, patch state
- **ITSM integration**: Incident history, SLA context, change records
- **Asset metadata**: Ownership, criticality, lifecycle stage

Correlation across sources enables anomaly detection that no single tool could achieve alone.

LAYER 2

RAG: Enterprise Knowledge, On Demand

Retrieval-Augmented Generation (RAG) gives agents access to institutional knowledge at inference time without baking it into model weights.



- **Knowledge Sources**

  Runbooks, KB articles, incident postmortems, vendor docs

- **Contextual Retrieval**

  Agent queries return the most relevant procedural context for each situation

- **Continuous Ingestion**

  New resolutions and policies update the knowledge base automatically

LAYER 3

Six Specialized Agents

Rather than one general-purpose agent, ADOF distributes responsibility across six domain-specific agents each with its own data feeds, tools, and escalation logic.



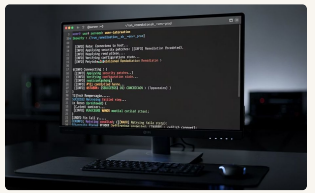
Incident Prediction

Detects precursor signals before tickets are created



Asset Health

Monitors device lifecycle and predicts hardware failure



Auto-Remediation

Executes approved fixes for repeatable, low-risk incidents



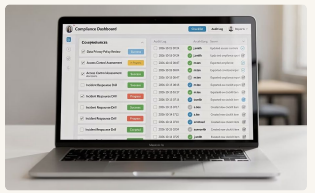
Procurement Intelligence

Triggers replacement or refresh workflows based on asset signals



User Experience

Correlates UX degradation signals with infrastructure events



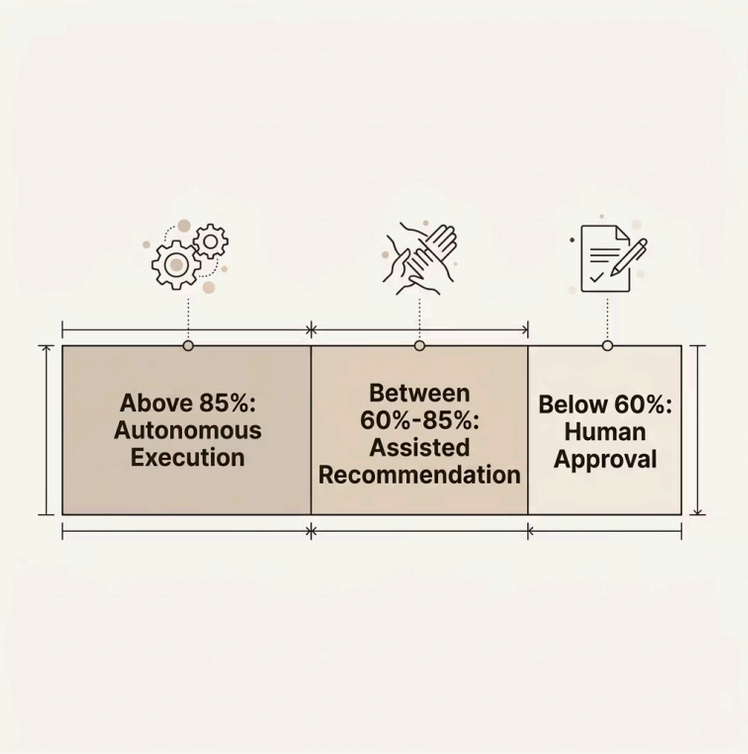
Compliance & Audit

Validates all autonomous actions against policy guardrails

LAYER 4 — CORE DESIGN DECISION

The Confidence Gate

Before any agent executes an action, a **composite confidence score** is calculated across three dimensions. This score governs whether the platform acts autonomously or routes for human approval.



Enterprise Health Score

System stability and blast radius assessment

Business Impact Score

Criticality of affected users, services, and SLAs

Automation Confidence Score

Historical success rate for this action pattern

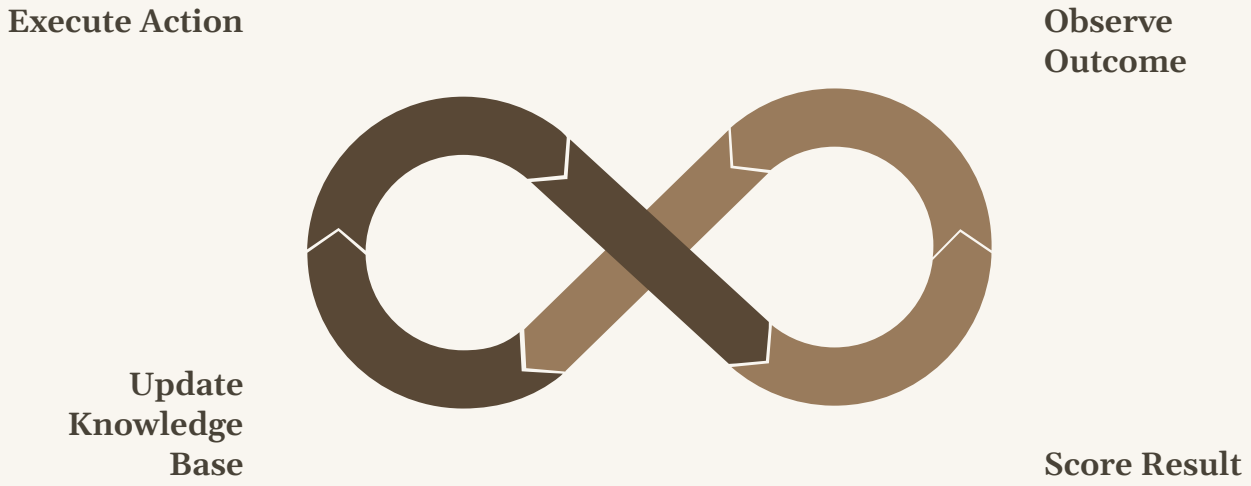
Above 85%: Execute.

Below 60%: Route to a human. The confidence gate is the difference between autonomous speed and governed safety and it makes both possible in the same platform.



LAYER 5 & 6

Execution, Feedback, and the Learning Loop



Every action ADOF takes feeds back into the system. Outcomes update the RAG knowledge base, refine confidence thresholds, and improve future agent decisions.

- Successful remediations reinforce automation confidence scores
- Human overrides are captured as labeled training signal
- Drift detection prevents model degradation over time

The system gets smarter with every incident not just faster.

GOVERNANCE MODEL

Five-Level Autonomy Maturity Model

1

Level 1 — Observe

Agents monitor and alert. No automated actions.

2

Level 2 — Recommend

Agents surface fix options; humans approve each action.

3

Level 3 — Assisted Execution

Low-risk, high-confidence actions execute automatically.

4

Level 4 — Supervised Autonomy

Broad execution with exception-based human review.

5

Level 5 — Full Autonomy

Self-healing with continuous policy validation and audit logging.



ILLUSTRATIVE TARGETS

What ADOF Is Designed to Deliver

40%

Level-1 Reduction

Projected decrease in repetitive support requests

85%

Confidence Threshold

Minimum score for fully autonomous remediation

6

Specialized Agents

Domain-specific agents with divided responsibility

5

Maturity Levels

Staged autonomy expansion with governance at each level

Targets are illustrative and proposed for future validation. Actual outcomes depend on organizational environment, data quality, and deployment scope.

Design Patterns You Can Apply Today



1 Unify Before You Automate

Signal correlation is the prerequisite. Don't build agents on siloed data.

2 Gate on Confidence, Not Rules

Static rule-based automation breaks. Composite scoring adapts.

3 Specialize Your Agents

Domain-specific agents outperform general-purpose ones in production.

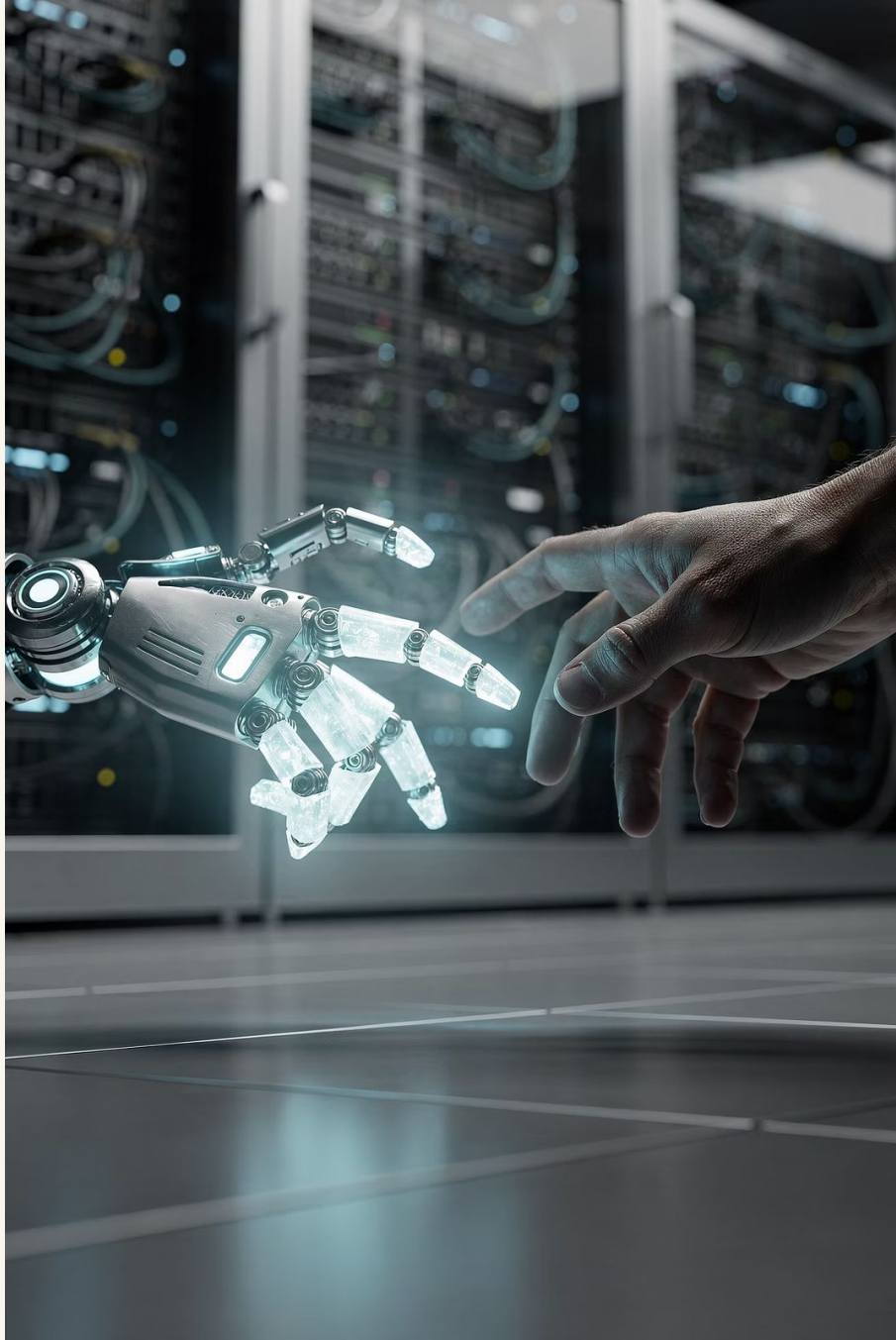
4 Build the Feedback Loop First

Without outcome capture, your system can't improve. Make it structural, not optional.

The Path to Self-Healing Operations

ADOF is a reference architecture not a finished product. The goal is to give teams a governed, composable blueprint for moving from reactive ops to proactive, autonomous platforms at their own pace, with human trust built in from the start.

Start with Signal Unification	**Gate Every Action on Confidence**
Expand Autonomy Incrementally	



Thank You!

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